

1. (Fall 2024, #2) Compute the indefinite integrals.

(a) $\int (x^2 - x + 1)e^x dx$

(b) $\int \sin^3 x dx$

2. (Fall 2023, #2) Compute the following antiderivatives.

(a) $\int x(x^2 + 1) \ln(x^2 + 1) dx$

(b) $\int \frac{\sqrt{t^2 - 9}}{t^6} dt$

3. (Spring 2023, #4) Evaluate $\int \frac{x^2}{(\sqrt{1-4x^2})^3} dx$.

4. (Fall 2022, #2) Compute the following indefinite and definite integrals:

(a) $\int x(\ln x)^2 dx$

(b) $\int_{-1}^1 (1-x)\sqrt{1-x^2} dx$

5. (Fall 2020, #2) Evaluate the following integrals.

(a) $\int e^{1-ax} \sin(1+ax) dx$ where a is any real number.

(b) $\int \frac{x}{\sqrt{4x^2 + 4x + 5}} dx$